AvnetAvenue™

Denali RLDRAM IP Core Demonstration Kit

The Denali RLDRAM controller IP Core is a critical time saving piece when designing with Reduced Latency DRAM. The Denali IP Core is a fully functional memory controller that generates the necessary commands needed to utilize the functionality of the RLDRAM. With the information provided, the IP Core can be implemented effectively into your Xilinx Virtex™-II FPGA through the IP Cores interface signals. The Denali RLDRAM controller IP Core provides you with everything needed to successfully implement a proven RLDRAM 400MHz (200MHz DDR) design (3200MB/sec peak). Designed with the Denali Databahn Memory Processor, the controller automatically maps the incoming user address to the available DRAM memory connected to the controller in a contiguous block.

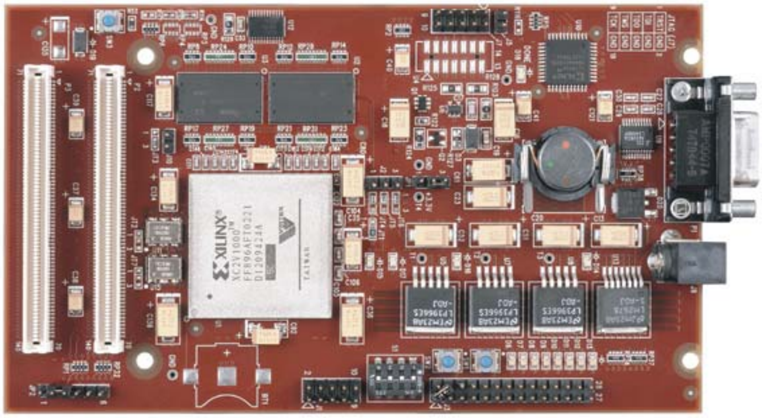
The IP Core is equipped with I/O cell circuitry used for compensation to meet timing requirements, such as: programmable read clock delay and programmable write clock delay. The IP Core has two separate read clocks, each driven by a DQS signal from a memory device. DDR I/O cells in conjunction with the on board DCMs of the FPGA maximize the data valid window to reliably capture and send data to and from the RLDRAM devices. ECC and BIST circuitry are also ready to be implemented if necessary.

The Denali RLDRAM IP Core Demonstration Kit includes:

- Design files
  - Denali RLDRAM IP Core EDIF Netlist
  - UCF (constraints) file (Xilinx XC2V1000-FF896 targeted)
  - Test bench
  - Example design
- Demonstration board
  - Command line controlled user test/interface
- Documentation
  - Denali RLDRAM IP Core data sheet
  - User's guide
  - Device porting guide
  - Schematic
  - Bill of materials
  - Hardware routing guidelines

Features:

- Controller IP Core
  - Pipelined queue based interface
  - Single port queued read/write interface
  - Register interface
  - Reset and initialization functions
  - Self refresh and power down commands
- RLDRAM
  - Two 8M X 32 RLDRAMs (512Megabits total) (Infineon - HYB18RL25632AC-5)
  - Memory bus speed of 400MHz (200MHz DDR)
- FPGA
  - Xilinx® Virtex-II XC2V1000-6 - FF896AFT0145
  - Xilinx XC18V04VQ44C
- Board I/O connectors
  - Two 140-pin general purpose I/O expansion connectors (AvBus)
- Power
  - Supplied AC/DC +12.0 V@3A power supply
- Communication
  - Easy to use command line user interface
  - RS-232 serial port (DB9)
- Configuration
  - JTAG header connector
  - In-system programmable PROM
  - 4 DIP-switches, 2 push-buttons



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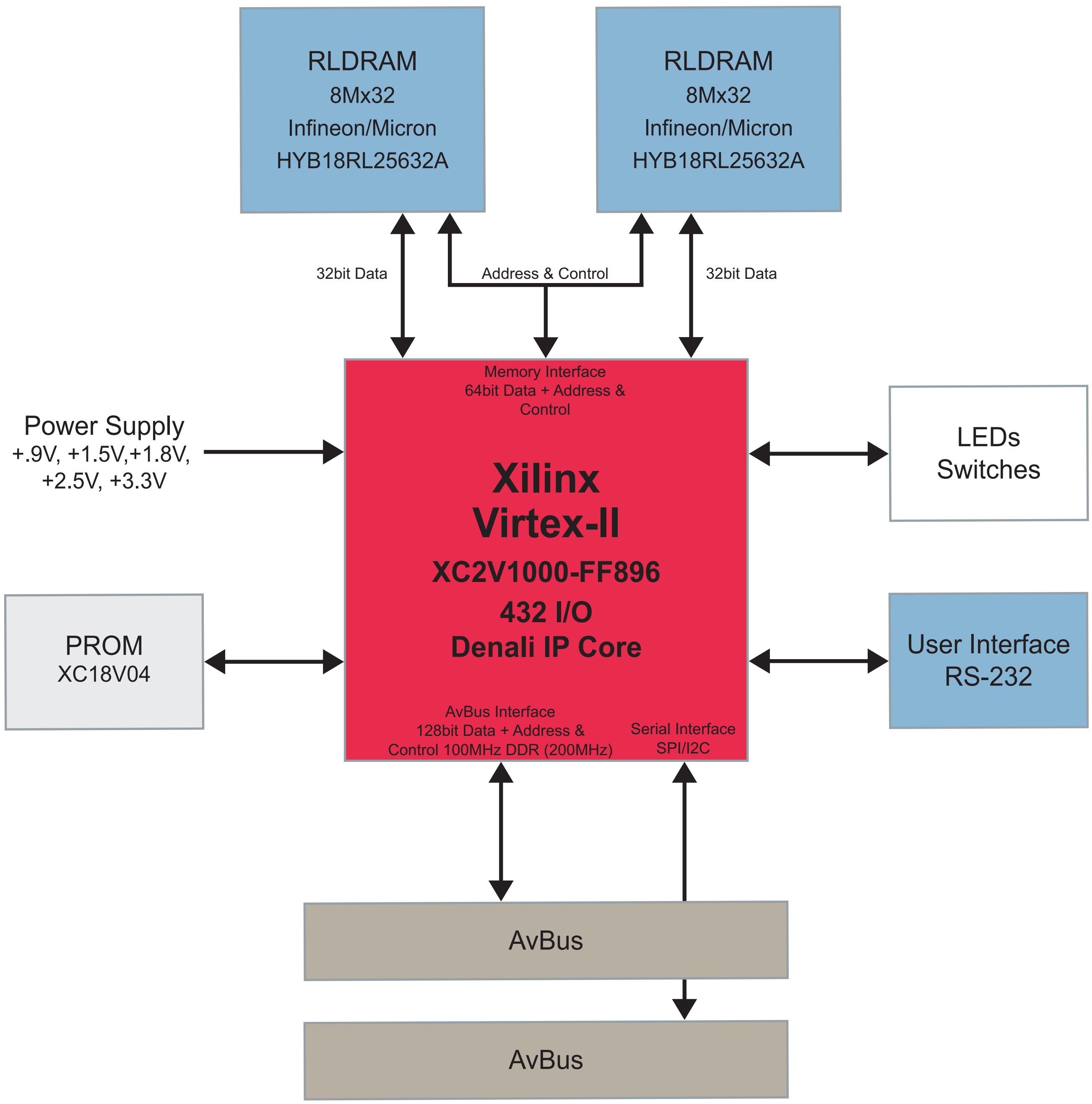

Supporting suppliers: Denali (Memory solutions defined.), Infineon technologies, Micron (The future of memory), XILINX®

Ordering Information:

Part Number	Hardware	Resale
ADS-DLI-RLDRAM	Denali core bundled with RLDRAM daughter card	$20,000.00 USD

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